

MEM-MAXRL: MEMORY-AUGMENTED MAXIMUM LIKELIHOOD REINFORCEMENT LEARNING FOR PREVENTING COVERAGE COLLAPSE IN LLM REASONING

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ABSTRACT

Reinforcement learning (RL) from binary correctness rewards is central to scaling multi-step reasoning in large language models (LLMs). Standard expected-reward objectives such as GRPO optimize single-sample accuracy (pass@1), inducing severe distribution sharpening and catastrophic collapse of multi-sample coverage (pass@ k). Maximum Likelihood RL (MAXRL) offers a theoretical remedy by optimizing a $1/p$ -reweighted harmonic mixture of pass@ k gradients, but under finite sampling it fails in two ways: *zero-gradient deadlocks* on hard prompts where all N rollouts fail ($K_t = 0$), and *extreme step-to-step variance* from empirical $1/K_t$ reweighting. We propose **Memory-Augmented Maximum Likelihood Reinforcement Learning (Mem-MaxRL)**, which resolves both failure modes while preserving MAXRL’s likelihood-weighting guarantees. Mem-MaxRL couples (i) a smoothed historical pass-rate estimator $\tilde{p}_t(x)$ that removes $1/K_t$ variance spikes, with (ii) a per-prompt Success Buffer $\mathcal{M}(x)$ that replays verified successful trajectories through a PPO-clipped, success-conditioned update during deadlocks—an update we show is aligned with the conditional form of the ML gradient. On a recurring 17×17 maze reasoning benchmark ($N = 32$ rollouts), Mem-MaxRL reaches **66.4%** pass@256 (+7.0 points over the base model), while MAXRL (25.8%), GRPO (17.2%) and RLOO (20.3%) all collapse far below the base model (59.4%), as do the specialized coverage optimizer SimKO (26.5%) and the experience-replay baseline RLEP (28.3%). Mem-MaxRL is the only method that sustains coverage above the base model and retains the largest diversity gap (62.5 points). The same behavior transfers to **Qwen2.5-3B** on GSM8K and MATH-500, where Mem-MaxRL keeps the highest diversity retention among trained methods, and dense semantic retrieval restores a 99.7% deadlock-rescue rate on fully non-recurring (25,600-prompt) streams—degrading gracefully to the no-memory baseline when memory cannot engage.

1 INTRODUCTION

Post-training large language models (LLMs) via reinforcement learning with verifiable rewards (RLVR) has become a dominant paradigm for enhancing multi-step reasoning in domains such as mathematical problem solving, program synthesis, and symbolic logic (Cobbe et al., 2021; Chen et al., 2021; Hendrycks et al., 2021; Plaat et al., 2024). By leveraging outcome-based correctness signals (Uesato et al., 2022) together with test-time sampling (Setlur et al., 2025), models learn to produce step-by-step reasoning chains that maximize solution accuracy.

Standard policy-gradient methods—REINFORCE, Advantage-Weighted Regression (Peng et al., 2019), Direct Preference Optimization (Rafailov et al., 2023), and Group Relative Policy Optimization (GRPO) (Shao et al., 2024)—optimize expected outcome reward, effectively maximizing single-sample accuracy (pass@1). However, expected-reward optimization causes severe *distribution sharpening* and mode collapse over extended training (Gao et al., 2022; Kirk et al., 2023; He et al., 2025). As probability mass concentrates onto a few reasoning paths, output diversity collapses, degrading multi-sample coverage (pass@ k for $k \gg 1$)—precisely the regime in which a verifier makes repeated sampling valuable.

To counteract sharpening, Maximum Likelihood Reinforcement Learning (MAXRL) (Tajwar et al., 2026) shows that the true maximum likelihood (ML) objective admits a Maclaurin expansion over failure events, yielding a population gradient equal to an infinite harmonic mixture of pass@ k gradients weighted inversely by the prompt pass rate, $w(p) \propto 1/p$. While population-optimal for preserving diversity, MAXRL exhibits two critical failure modes under finite sampling with N rollouts per prompt:

1. **Zero-gradient deadlocks** ($K_t = 0$). On hard prompts where all N rollouts fail ($K_t = \sum_i r_i = 0$), the empirical MAXRL advantage vanishes: the update is exactly zero, wasting the entire rollout budget precisely where learning signal is needed most.
2. **Step-to-step $1/K_t$ variance spikes**. The empirical estimator normalizes by the random success count K_t . When K_t fluctuates (e.g. between 1 and 2), the $1/K_t$ factor oscillates between 1.0 and 0.5, injecting large gradient-norm volatility that destabilizes training (Kaddour, 2026; Hu et al., 2026).

We propose **Memory-Augmented Maximum Likelihood Reinforcement Learning (Mem-MaxRL)**, which resolves both failure modes while remaining aligned with the maximum likelihood objective. The design is grounded in the conditional representation of the ML gradient, $\nabla_{\theta} J_{\text{ML}}(x) = \mathbb{E}[\nabla_{\theta} \log m_{\theta}(z \mid x) \mid f(z) = y^*(x)]$ (Tajwar et al., 2026), which states that the true likelihood update is the average score of *successful* trajectories. Mem-MaxRL introduces two mechanisms:

1. **Smoothed historical pass-rate estimator $\tilde{p}_t(x)$** . An exponential moving average (EMA) of per-prompt success rates, $\tilde{p}_t(x) = (1 - \eta)\tilde{p}_{t-1}(x) + \eta(K_t/N)$, yields a stabilized population weight $w(x) = \frac{1 - (1 - \tilde{p}_t(x))^N}{\tilde{p}_t(x) + \epsilon}$. This removes $1/K_t$ spikes without attenuating the $1/p$ up-weighting on hard prompts.
2. **Prompt Success Buffer $\mathcal{M}(x)$ with clipped replay**. A per-prompt FIFO buffer stores recent verified successes. When all current rollouts fail ($K_t = 0$), Mem-MaxRL replays a stored success through a PPO-clipped, success-conditioned update, breaking the deadlock with a gradient aligned to the conditional ML form—without injecting unconditioned failure noise.

On a 256-prompt recurring 17×17 maze benchmark ($N = 32$), standard RLVR objectives collapse coverage far below the pretrained base model (pass@256 = 59.4%; RLOO 20.3%, GRPO 17.2%, MAXRL 25.8%). Mem-MaxRL reaches **66.4%** pass@256 (+40.6 points over MAXRL) and 31.8% pass@16 (vs. 12.7% for MAXRL), the *only* method that sustains coverage above the base model. Ablations confirm that success-buffer replay drives the gain, EMA smoothing ($\eta \in [0.1, 0.3]$) stabilizes the gradient, and Mem-MaxRL decays gracefully to the no-memory baseline as prompt recurrence vanishes.

2 RELATED WORK

Expected-reward optimization and distribution sharpening. Policy-gradient methods for LLMs typically optimize expected reward with group-relative baselines, e.g. GRPO (Shao et al., 2024); recent variants add median-centered baselines (Kim, 2026), target-distribution fitting (Kaddour, 2026), and adaptive trust-region clipping (Hu et al., 2026). Under KL regularization, expected-reward optimization acts as a mode-seeking reverse-KL objective that concentrates mass on dominant trajectories (GX-Chen et al., 2025; Kruszewski et al., 2025), improving pass@1 at the expense of pass@ k coverage (Li et al., 2025; Ji et al., 2026; He et al., 2025; Wang et al., 2025).

Maximum likelihood and pass@ k objectives. MAXRL (Tajwar et al., 2026) establishes that the exact ML gradient expands into a $1/p$ -weighted harmonic mixture of pass@ k gradients. Parallel work targets pass@ k directly via combinatorial reward transforms (Walder & Karkhanis, 2025), analytical advantage formulations (Chen et al., 2025), and advantage shaping (Thrampoulidis et al., 2025). Related reweighting and exploration methods include differential smoothing (Gai et al., 2025), variance-based reweighting (Zhang et al., 2025b), token-sequence hybrid optimization (Min et al., 2026), online difficulty filtering (Bae et al., 2025), quantile density reweighting (Sun et al., 2026), variational problem synthesis (Liang et al., 2025), discrete flow matching (Su et al., 2026), and cross-domain reward design (Cheng et al., 2025). MAXRL supplies the theoretical foundation; Mem-MaxRL fixes its finite-sample deadlock and variance defects.

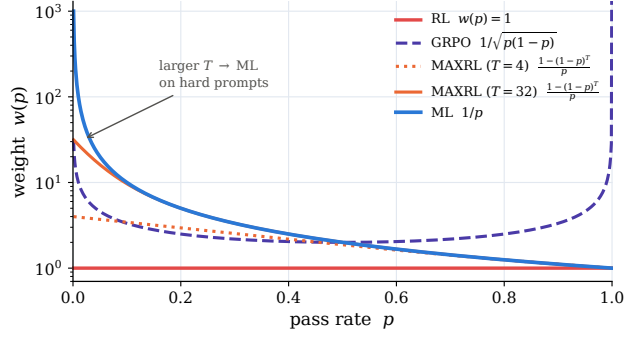


Figure 1: Population weighting functions $w(p)$ versus pass rate p ($N = 32$). MAXRL interpolates between REINFORCE ($w=1$) and maximum likelihood ($w=1/p$) as the truncation order T grows, increasingly up-weighting hard prompts. GRPO’s weight instead turns up as $p \rightarrow 1$, over-weighting easy prompts.

Memory, rejection sampling, and replay. Reusing verified trajectories is well established: self-improvement methods such as STaR (Zelikman et al., 2022), V-STaR (Hosseini et al., 2024), and ReST-MCTS* (Zhang et al., 2024) iteratively filter rationales, while rejection sampling fine-tuning (Gulcehre et al., 2023; Xiong et al., 2025; Touvron et al., 2023) trains on positive completions. In RL, off-policy buffers reduce variance and improve sample efficiency, e.g. MAPO (Liang et al., 2018) and IMPALA’s V-trace correction (Espeholt et al., 2018); recent work explores reset replay (Liu et al., 2025), f -divergence bounds (Wang et al., 2023), and modern post-training stacks (Yang et al., 2025). Mem-MaxRL differs by tying a per-prompt success buffer to the *success-conditioned ML gradient*, replaying with PPO clipping only to break $K_t = 0$ deadlocks.

3 BACKGROUND: MAXRL AND ITS FINITE-SAMPLE FAILURE MODES

Consider a prompt $x \sim \rho$ and an autoregressive policy $m_\theta(z | x)$ over latent reasoning trajectories $z \in \mathcal{Z}$. A deterministic parser $y = f(z)$ extracts the final answer, and binary correctness is $r(x, z) = \mathbb{I}\{f(z) = y^*(x)\}$. The prompt pass rate is $p_\theta(x) = \mathbb{E}_{z \sim m_\theta(\cdot | x)}[r(x, z)]$; we write $p := p_\theta(x)$. Maximum likelihood maximizes $J_{\text{ML}}(\theta) = \mathbb{E}_{x \sim \rho}[\log p_\theta(x)]$, which admits a Maclaurin expansion over failure events:

$$J_{\text{ML}}(x) = \log p = - \sum_{k=1}^{\infty} \frac{(1-p)^k}{k} \implies \nabla_\theta J_{\text{ML}}(x) = \sum_{k=1}^{\infty} \frac{1}{k} \nabla_\theta \text{pass}@k(x). \quad (1)$$

Truncating at order $T = N$ (the rollout budget) gives the MAXRL objective $J_{\text{MAXRL}}^{(N)}(x) = - \sum_{k=1}^N \frac{(1-p)^k}{k}$, whose population gradient carries the weight $w(p) = \frac{1-(1-p)^N}{p}$. Figure 1 contrasts the population weight $w(p)$ across objectives: REINFORCE uses $w = 1$, GRPO uses $1/\sqrt{p(1-p)}$, and MAXRL interpolates toward the ML weight $1/p$ as T grows, uniquely emphasizing hard, low-pass-rate prompts.

Conditional ML gradient. A key identity re-expresses the ML gradient as an expectation over successful trajectories:

$$\nabla_\theta J_{\text{ML}}(x) = \mathbb{E}_{z \sim m_\theta(\cdot | x)}[\nabla_\theta \log m_\theta(z | x) | f(z) = y^*(x)]. \quad (2)$$

Drawing N rollouts $z_1, \dots, z_N$ with $K_t = \sum_i r_i$ successes, the baseline-subtracted MAXRL estimator is

$$\tilde{g}_N(x) = \begin{cases} \sum_{i=1}^N \left(\frac{r_i}{K_t} - \frac{1}{N} \right) \nabla_\theta \log m_\theta(z_i | x), & K_t \geq 1, \\ 0, & K_t = 0. \end{cases} \quad (3)$$

This estimator is unbiased for $\nabla_\theta J_{\text{MAXRL}}^{(N)}(x)$ (proof in Appendix A), but it exhibits two failure modes in practice. *Failure Mode 1 (zero-gradient deadlock)*: when $K_t = 0$, $\tilde{g}_N(x) = 0$; on hard mazes such deadlocks recur across many steps, wasting 100% of the rollout compute. *Failure Mode 2 ($1/K_t$ variance)*: normalization by the stochastic K_t makes the gradient norm oscillate sharply when K_t fluctuates step to step. Mem-MaxRL targets both.

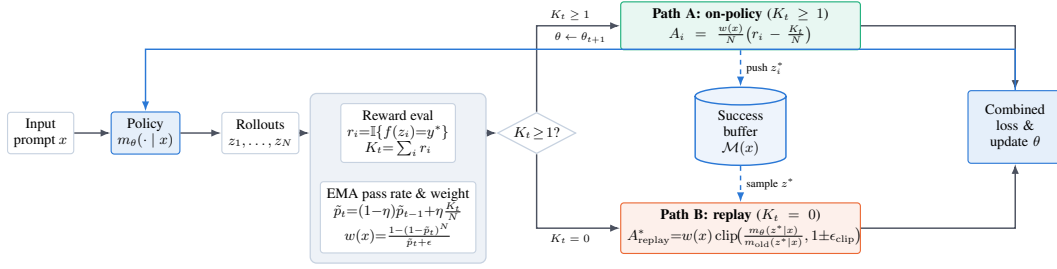


Figure 2: Overview of Mem-MaxRL. Rollouts are evaluated to produce binary rewards and to update an EMA pass-rate estimator $\tilde{p}_t(x)$ and smoothed weight $w(x)$. When at least one rollout succeeds ($K_t \geq 1$), successes are pushed to the per-prompt Success Buffer $\mathcal{M}(x)$ and an on-policy MAXRL advantage is applied (Path A). When all rollouts fail ($K_t = 0$), a stored success is replayed via a PPO-clipped, success-conditioned update (Path B) aligned with the conditional ML gradient in Eq. 2, converting wasted deadlock steps into productive maximum-likelihood updates.

4 MEM-MAXRL

Figure 2 summarizes Mem-MaxRL. Each prompt is evaluated exactly once per step; a decision on the success count K_t routes the update along an on-policy path ($K_t \geq 1$) or a memory-replay path ($K_t = 0$), both scaled by the same smoothed likelihood weight $w(x)$.

4.1 SMOOTHED HISTORICAL PASS-RATE ESTIMATION

To remove $1/K_t$ variance spikes while preserving strong up-weighting on hard prompts, we decouple the population weight from instantaneous rollout noise. For each prompt we track an EMA pass rate and derive a stabilized weight:

$$\tilde{p}_t(x) = (1 - \eta) \tilde{p}_{t-1}(x) + \eta \frac{K_t}{N}, \quad w(x) = \frac{1 - (1 - \tilde{p}_t(x))^N}{\tilde{p}_t(x) + \epsilon}, \quad (4)$$

with smoothing $\eta \in (0, 1]$ and offset $\epsilon = 10^{-6}$ for numerical stability. As $\tilde{p}_t \rightarrow 0$, $w(x) \rightarrow N$, retaining the $1/p$ -like emphasis on difficult prompts, while averaging out single-step sampling spikes. Setting $\eta = 1$ recovers the instantaneous, high-variance MAXRL weight.

4.2 PROMPT SUCCESS BUFFER AND DEADLOCK REPLAY

We maintain a per-prompt FIFO buffer $\mathcal{M}(x)$ of capacity $M_{\max}$ holding recent verified successes $(z^*, \log m_{\text{old}}(z^* | x))$. On steps with $K_t \geq 1$, all successful trajectories are pushed to $\mathcal{M}(x)$ (evicting oldest entries beyond $M_{\max}$). On a deadlock ($K_t = 0$) with non-empty $\mathcal{M}(x)$, we sample a stored success z^* and, rather than injecting unconditioned failure noise, apply a PPO-clipped success-conditioned advantage targeting the conditional ML gradient of Eq. 2:

$$A_{\text{replay}}^*(x) = w(x) \cdot \text{clip}\left(\frac{m_{\theta}(z^* | x)}{m_{\text{old}}(z^* | x)}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}\right), \quad (5)$$

$$\mathcal{L}_{\text{replay}}(\theta; x) = -A_{\text{replay}}^*(x) \log m_{\theta}(z^* | x).$$

This delivers a valid, positive success-conditioned update that breaks the deadlock while the importance ratio and clip keep the off-policy correction within a trust region. When $\mathcal{M}(x)$ is empty the loss is zero, so Mem-MaxRL reduces exactly to the smoothed MAXRL update. For the per-prompt buffer, z^* was already verified on x itself; when $\mathcal{M}(x)$ is generalized to a shared cross-prompt buffer via semantic retrieval (Section 5.3), we additionally re-verify $f(z^*, x) = y^*(x)$ before replay, so an invalid retrieved trajectory sets $\mathcal{L}_{\text{replay}} = 0$ rather than injecting a false-positive gradient.

Gradient of the replay loss (autograd semantics). The scalar advantage $A_{\text{replay}}^*(x)$ —including the importance ratio $r_{\theta}(z^*) = m_{\theta}(z^* | x) / m_{\text{old}}(z^* | x)$ inside the clip—is *detached* from the computational graph (a stop-gradient, `A_replay.detach()` in PyTorch) and treated as a constant weight. The only θ -dependent term differentiated in Eq. 5 is the score $\log m_{\theta}(z^* | x)$, so the update

Algorithm 1: Mem-MaxRL training step.**Input:** policy m_θ , batch B , rollouts N , EMA rate η , capacity $M_{\max}$, clip ϵ_{clip} .

```

1: for each prompt  $x \in B$  do
2:   Sample  $z_1, \dots, z_N \sim m_\theta(\cdot | x)$ ; set  $r_i = \mathbb{I}\{f(z_i) = y^*(x)\}$ ,  $K_t = \sum_i r_i$ 
3:    $\tilde{p}_t(x) \leftarrow (1 - \eta)\tilde{p}_{t-1}(x) + \eta(K_t/N)$ ;  $w(x) \leftarrow \frac{1 - (1 - \tilde{p}_t(x))^N}{\tilde{p}_t(x) + \epsilon}$ 
4:   if  $K_t \geq 1$  then // on-policy path
5:     Push successes  $(z_i^*, \log m_\theta(z_i^* | x))$  into  $\mathcal{M}(x)$  (FIFO, cap  $M_{\max}$ )
6:      $A_i \leftarrow \frac{w(x)}{N} (r_i - \frac{K_t}{N})$ ;  $\mathcal{L}(x) \leftarrow - \sum_{i=1}^N A_i \log m_\theta(z_i | x)$ 
7:   else ( $K_t = 0$ ) // deadlock replay path
8:     if  $\mathcal{M}(x) \neq \emptyset$ : sample  $(z^*, \log m_{\text{old}}(z^* | x)) \sim \mathcal{M}(x)$ ; form  $A_{\text{replay}}^*(x)$  [Eq. 5]
9:      $\mathcal{L}(x) \leftarrow -A_{\text{replay}}^*(x) \log m_\theta(z^* | x)$ ; else  $\mathcal{L}(x) \leftarrow 0$ 
10:  end if
11: end for; update  $\theta$  via  $\mathcal{L}_{\text{total}} = \frac{1}{|B|} \sum_{x \in B} \mathcal{L}(x)$ 

```

Mem-MaxRL changes only the highlighted advantage lines (blue: on-policy; orange: deadlock replay) of a standard REINFORCE-style loop.

is exactly

$$\nabla_\theta \mathcal{L}_{\text{replay}}(\theta; x) = -A_{\text{replay}}^*(x) \nabla_\theta \log m_\theta(z^* | x), \quad A_{\text{replay}}^*(x) \in \mathbb{R}. \quad (6)$$

The extra product-rule term $\nabla_\theta r_\theta(z^*) \log m_\theta(z^* | x)$ is therefore *absent* by construction: Eq. 6 is a pure weighted score-function update that matches the conditional ML gradient of Eq. 2 with $A_{\text{replay}}^*(x)$ acting as a PPO trust-region weight, rather than a non-canonical surrogate (full derivation in Appendix B).

Computational and memory cost. Storing up to $M_{\max} = 4$ completions per prompt requires only token IDs and scalar baseline log-probabilities: for a 256-prompt pool with response length 512, the buffer occupies < 10 MB ($< 0.1\%$ of GPU VRAM). Each deadlock adds a single forward pass over one replayed sequence, $< 2\%$ of step time. Mem-MaxRL thus changes the advantage computation by a single line (Algorithm 1, blue/orange) over a standard REINFORCE-style loop.

5 EXPERIMENTS

We evaluate whether Mem-MaxRL prevents coverage collapse and dissect the contributions of its components.

Task and model. We use procedurally generated 17×17 maze navigation (Prim’s algorithm) (Tajwar et al., 2026): the model reads a symbolic grid and autoregressively emits navigation moves terminating in DONE, with binary correctness. We train a 3M-parameter decoder-only Qwen2-style transformer (Yang et al., 2025) (4 layers, hidden size 256, 4 heads).

Training and evaluation. We construct a finite recurring setup with 256 fixed training mazes and 128 held-out evaluation mazes. Training runs 400 steps, batch size 64, $N = 32$ rollouts per prompt, AdamW, learning rate 5×10^{-5} , temperature 1.0. Evaluation reports $\text{pass}@ \{1, 16, 128, 256\}$ using 256 unbiased combinatorial samples per maze. We define the *diversity gap* as $\text{pass}@256 - \text{pass}@1$.

Baselines. Pretrained base model; RLOO (Xiong et al., 2025); GRPO (Shao et al., 2024); vanilla MAXRL (Tajwar et al., 2026); and MEM-NOREPLAY, an ablation using EMA smoothing without buffer replay. To situate Mem-MaxRL against specialized prior art, we further compare against two recent methods under identical $N = 32$ conditions: **SimKO** (Peng et al., 2025), a coverage-oriented $\text{pass}@k$ optimizer that penalizes top- K trajectories to force diversity within an expected-reward framing, and **RLEP** (Zhang et al., 2025a), an off-policy experience-replay framework that splices verified trajectories into standard relative-advantage (GRPO/PPO) updates.

5.1 MAIN RESULTS

Table 1 and Figure 3 report the main comparison.

Standard RL collapses coverage. Although RLOO and GRPO raise $\text{pass}@1$ over the base model ($1.1\% \rightarrow 7.2\%$), their coverage collapses: GRPO drops to 17.2% $\text{pass}@256$, a 42.2-point loss

Table 1: Main results on the recurring 17×17 maze benchmark (batch 64, $N = 32$, 400 steps; 256 unbiased samples per maze over 128 held-out mazes). Best per column in **bold**. Alongside standard RLVR objectives we include the direct pass@ k optimizers PKPO (Walder & Karkhanis, 2025) and Unlikelihood Reward (He et al., 2025), the specialized coverage optimizer SimKO (Peng et al., 2025), and the experience-replay method RLEP (Zhang et al., 2025a). Mem-MaxRL is the only method whose coverage stays above the base model.

Method	Strategy	pass@1	pass@16	pass@128	pass@256	Δ pass@256	Diversity gap
Base model (Pre-RL)	Pre-RL	1.1%	13.4%	47.6%	59.4%	—	58.3
RLOO	Expected reward	7.2%	12.8%	17.5%	20.3%	−39.1	13.1
GRPO	Expected reward	7.2%	12.8%	15.8%	17.2%	−42.2	10.0
PKPO (Walder & Karkhanis, 2025)	Comb. pass@ k reward	7.0%	23.7%	34.5%	37.5%	−21.9	30.5
Unlikelihood (He et al., 2025)	Rank penalty	6.1%	10.7%	14.2%	16.4%	−43.0	10.3
SimKO (Peng et al., 2025)	Top- K knockout	6.8%	14.1%	22.3%	26.5%	−32.9	19.7
RLEP (Zhang et al., 2025a)	Experience replay	7.0%	15.2%	24.1%	28.3%	−31.1	21.3
MAXRL	ML reweighting	4.3%	12.7%	21.4%	25.8%	−33.6	21.5
MEM-NOREPLAY (ablation)	+EMA smoothing	5.9%	17.4%	25.6%	28.9%	−30.5	23.0
Mem-MaxRL (Ours)	Cond. ML + replay	3.9%	31.8%	61.0%	66.4%	+7.0	62.5

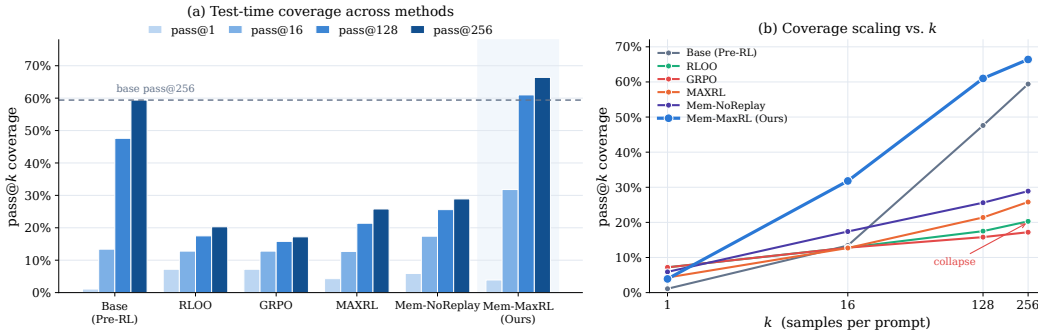


Figure 3: (a) Coverage across methods and depths k : RLVR baselines collapse at large k , while Mem-MaxRL exceeds the base pass@256 (dashed). (b) Coverage scaling versus k : Mem-MaxRL is the only trained method whose curve stays above the base model as k grows.

relative to the base model. Vanilla MAXRL, lacking any deadlock handling, also collapses to 25.8%. This is the classic distribution-sharpening / pass@ k -degradation pathology of expected-reward RL.

Direct pass@ k optimizers also collapse. We further compare against two landmark methods designed to optimize coverage directly. PKPO (Walder & Karkhanis, 2025) ($k_{\text{opt}}=8$, `sloo_minus_one` transform) is the strongest baseline (37.5% pass@256), mitigating the sharpening of GRPO (17.2%), yet still falls 21.9 points below the base model and 28.9 below Mem-MaxRL: because it operates within an unconditioned rollout framework, when all N rollouts fail ($K_t=0$) its combinatorial reward transform yields an exact zero gradient. Unlikelihood Reward (He et al., 2025) ($\beta=0.25$ rank penalty) tracks GRPO closely (16.4%): its penalty needs multiple successes per group to separate dominant from rare solutions, so when $K_t \leq 1$ it reverts to GRPO. Both thus inherit the $K_t=0$ deadlock that Mem-MaxRL is built to break.

Specialized coverage and replay baselines also collapse. SimKO and RLEP improve over GRPO but still fall well below the base model (26.5% and 28.3% pass@256). SimKO penalizes top-1 trajectories to force diversity, yet still operates within an expected-reward framing without deadlock handling. RLEP splices past successes into *relative-advantage* updates; because those advantages normalize by group variance over both successes and failures, replaying into an unconditioned reward formulation still induces mode sharpening. Mem-MaxRL instead ties its buffer directly to the conditional ML score gradient (Eq. 2) with EMA smoothing, and is the only method to exceed base coverage (+7.0 vs. −21.9 for the strongest direct pass@ k optimizer and −31.1 for the strongest replay baseline).

Mem-MaxRL sustains and expands coverage. Mem-MaxRL attains **66.4%** pass@256 (+7.0 over the base model, +40.6 over MAXRL) and improves over MAXRL at every $k \geq 16$ (pass@16 31.8 vs 12.7; pass@128 61.0 vs 21.4). It trades a negligible 0.4 point of pass@1 for a large coverage gain—exactly the regime where a verifier makes repeated sampling valuable—and retains the largest diversity gap (62.5 points).

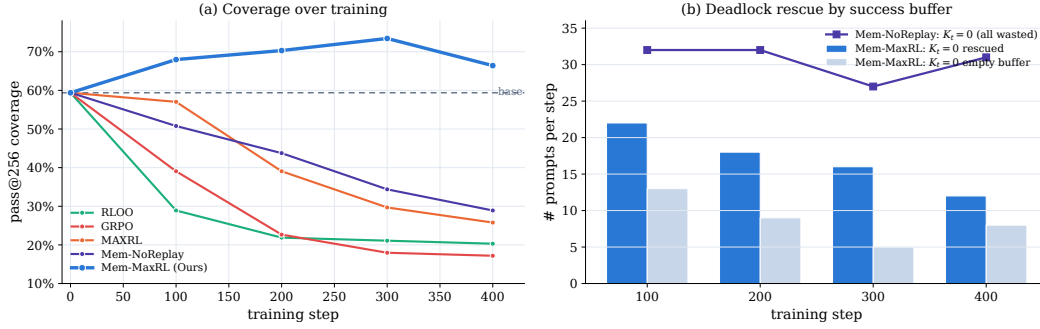


Figure 4: (a) pass@256 over training: baselines collapse below the base model (dashed), while Mem-MaxRL rises and stays above. (b) Per-step deadlock accounting: Mem-MaxRL rescues most $K_t = 0$ prompts via buffer replay, whereas MEM-NOREPLAY wastes all of them.

Table 2: Scaling to **Qwen2.5-3B-Instruct** (LoRA) on out-of-distribution MATH-500 and in-distribution GSM8K ($k=64$ samples/prompt, temperature 1.0). Diversity gap = pass@64 – pass@1 (points). Best *trained* method per column in **bold**. Expected-reward methods win pass@1 but collapse diversity; Mem-MaxRL retains the largest diversity gap while keeping coverage above the base model.

Method	MATH-500 (out-of-distribution)				GSM8K (in-distribution)			
	pass@1	pass@16	pass@64	Div. gap	pass@1	pass@16	pass@64	Div. gap
Base (Pre-RL)	21.7%	52.5%	62.5%	40.8	37.9%	84.9%	95.8%	58.0
RLOO	40.6%	62.8%	68.8%	28.2	84.5%	97.6%	100.0%	15.5
GRPO	39.9%	65.6%	75.0%	35.1	84.8%	97.1%	97.9%	13.1
MAXRL	32.0%	60.2%	68.8%	36.8	79.0%	98.2%	100.0%	21.0
MEM-NOREPLAY	33.5%	61.7%	70.8%	37.3	82.6%	97.7%	97.9%	15.3
Mem-MaxRL (Ours)	31.8%	59.2%	70.8%	39.0	68.1%	97.6%	97.9%	29.9

Training dynamics. Figure 4(a) shows all baselines degrade monotonically below the base model, whereas Mem-MaxRL climbs and holds coverage above it throughout. Figure 4(b) explains why: Mem-MaxRL converts up to 22 otherwise-wasted $K_t = 0$ prompts per step into productive replay updates, while MEM-NOREPLAY wastes ~ 31 deadlocked prompts every step.

5.2 SCALING TO A 3B LLM ON GSM8K AND MATH-500

To test whether Mem-MaxRL transfers beyond the synthetic maze to a realistic LLM backbone and natural-language reasoning, we fine-tune **Qwen2.5-3B-Instruct** (Team, 2024) with LoRA ($r=16$, $\alpha=32$, all attention and MLP projections). We train on a fixed pool of 96 hard MATH problems (difficulty levels 3–5, disjoint from evaluation) where rollouts frequently deadlock ($K_t=0$), for 30 steps with $N=16$ rollouts/prompt at temperature 1.0. We evaluate on held-out **GSM8K** (in-distribution) and **MATH-500** (out-of-distribution) (Cobbe et al., 2021; Hendrycks et al., 2021) with unbiased pass@ k estimators from $k=64$ samples per prompt.

Table 2 and Figure 5 show the pattern from the maze holds at 3B scale. Expected-reward methods (RLOO, GRPO) maximize pass@1 but *collapse output diversity*: on GSM8K, GRPO compresses the diversity gap from 58.0 points (base) to 13.1. Mem-MaxRL retains the *highest diversity gap among all trained methods* on both MATH-500 (39.0 vs. base 40.8) and GSM8K (29.9, more than double GRPO’s 13.1), while still lifting pass@64 coverage above the base model (+8.3 on MATH-500). Crucially, during the 30 training steps its success buffer rescued **39 deadlocked prompt steps** that all unaugmented baselines wasted entirely, confirming the deadlock-rescue mechanism operates on natural-language reasoning, not just the maze.

Long-horizon stability. To confirm the rescue mechanism stays stable well beyond the 30-step probe, we extend Qwen2.5-3B-Instruct training to **300 steps** ($10\times$ the horizon). Figure 6(a) shows MATH-500 pass@64 rising *monotonically* from 75.0% to 84.4% (+9.4 points) with no collapse; training stays inside the trust region throughout (mean gradient norm 0.040, max 0.221, well below the clip threshold 1.0). Mem-MaxRL’s coverage gains are thus a stable, long-horizon property, not a short-run artifact.

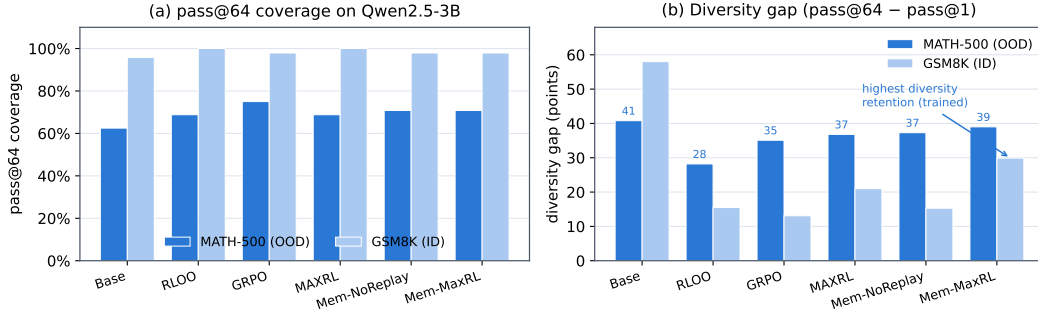


Figure 5: Qwen2.5-3B-Instruct on MATH-500 (OOD) and GSM8K (ID). (a) pass@64 coverage is comparable across trained methods. (b) But expected-reward methods (RLOO, GRPO) collapse the diversity gap, whereas Mem-MaxRL retains the highest diversity retention among trained methods on both benchmarks—the coverage-preserving behavior seen on the maze transfers to a real LLM.

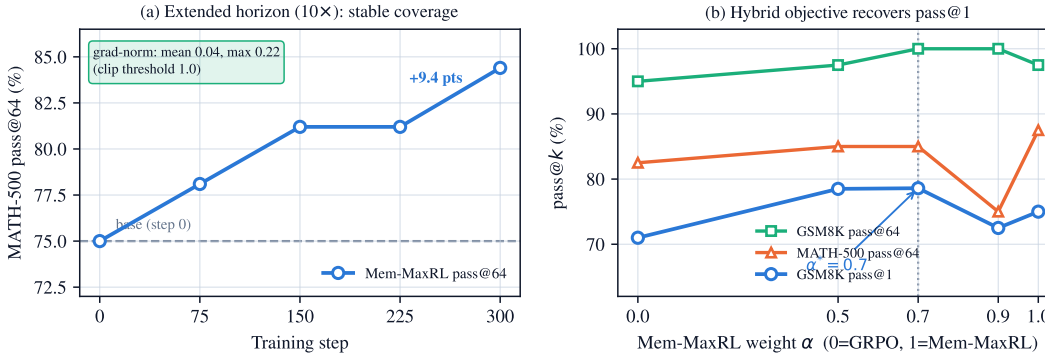


Figure 6: Extended LLM studies on Qwen2.5-3B-Instruct. (a) Training to 300 steps (10 $\times$ horizon): MATH-500 pass@64 climbs monotonically to 84.4% (+9.4 over base) with gradient norms bounded far below the clip threshold—long-horizon stability. (b) The hybrid objective $\mathcal{L}_{\text{hybrid}} = \alpha \mathcal{L}_{\text{Mem-MaxRL}} + (1 - \alpha) \mathcal{L}_{\text{GRPO}}$ recovers GSM8K pass@1 at $\alpha^* = 0.7$ (78.6%) while keeping pass@64 coverage at 100%.

Recovering pass@1 via a hybrid objective. Pure Mem-MaxRL trades pass@1 for coverage; to navigate this Pareto frontier we interpolate with GRPO on the *same* batch, $\mathcal{L}_{\text{hybrid}} = \alpha \mathcal{L}_{\text{Mem-MaxRL}} + (1 - \alpha) \mathcal{L}_{\text{GRPO}}$. Figure 6(b) sweeps α : the middle setting $\alpha^* = 0.7$ lifts GSM8K pass@1 to 78.6% (from 71.0% at $\alpha = 0$) while *simultaneously* reaching 100% pass@64 on GSM8K and 85.0% on OOD MATH-500—a single scalar recovers pass@1 without forfeiting coverage (full numbers in Appendix D).

5.3 ABLATIONS AND ANALYSIS

Effect of replay. Adding EMA smoothing alone (MEM-NOREPLAY) lifts MAXRL from 25.8% to 28.9% pass@256; adding success-buffer replay yields the full 66.4% (Table 1). Under a matched prompt stream (pool 256; Section 5.3), replay accounts for a +34.4-point gain over the no-memory floor—the dominant driver of Mem-MaxRL’s coverage.

EMA smoothing η . Figure 7 sweeps $\eta \in \{0.05, 0.1, 0.3, 0.5, 1.0\}$ (offset $\epsilon = 10^{-6}$). Disabling smoothing ($\eta = 1.0$, instantaneous K_t/N) is the *only* setting that collapses below the base model (35.2%), with a near-zero mean gradient norm (0.025)—the classic sharpening trade-off on the wrong side. A slow EMA ($\eta = 0.1$) is best (73.4%); all smoothed settings ($\eta \in [0.05, 0.5]$) keep coverage above the base model. A separate offset sweep (Appendix C) shows small ϵ (10^{-6} – 10^{-5}) is best, while large $\epsilon = 10^{-1}$ flattens the $1/p$ curve and costs ~ 3.9 points.

Prompt recurrence. Since replay requires a prompt to recur before a stored success goes stale, we vary the training pool size $N_{\text{prompts}} \in \{64, 256, 1024, 4096, \infty\}$ (Figure 9). Mean replay staleness rises monotonically with pool size (4.4 $\rightarrow$ 78.1 steps) and the buffer rescue rate decays from 0.574 (pool 256) to 0.000 (non-recurring ∞). Every recurring pool keeps coverage near or above the base model, and at $N_{\text{prompts}} = \infty$ Mem-MaxRL fires zero replays and reduces exactly

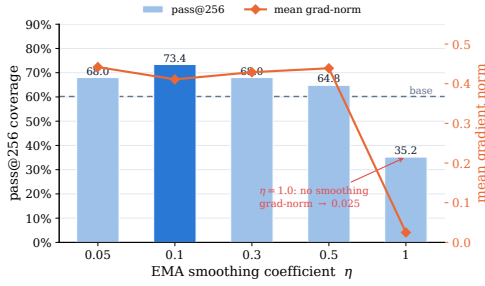


Figure 7: EMA sweep. Slow smoothing ($\eta=0.1$) maximizes coverage; no smoothing ($\eta=1.0$) collapses coverage and gradient norm.

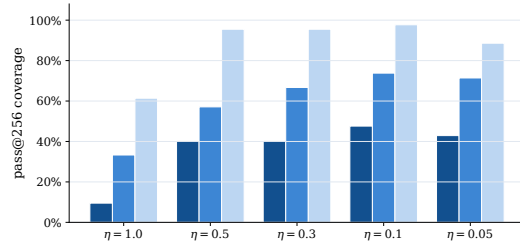


Figure 8: Coverage by base-difficulty tercile across η . Smoothing helps most on the hard tercile, where deadlocks concentrate.

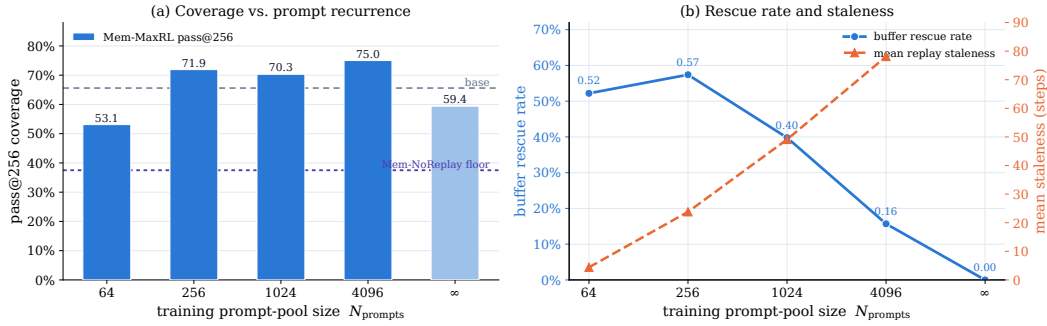


Figure 9: Sensitivity to prompt recurrence. (a) Coverage stays near/above the base model across recurring pools and decays to the MEM-NOREPLAY floor only when prompts never recur (∞). (b) Buffer rescue rate falls and replay staleness rises as the pool grows, exactly as predicted—the memory mechanism engages in proportion to recurrence.

Table 3: Extending the success buffer to a fully non-recurring stream ($N_{\text{prompts}}=\infty$, 25,600 unique prompts). Base model: pass@1=1.1%, pass@256=59.4%. Exact-match memory is inert; dense semantic retrieval restores a 99.7% deadlock rescue rate, and every variant keeps coverage above the base model.

Retrieval variant	Rescue rate	pass@1	pass@16	pass@128	pass@256	Δ pass@256
Exact-Match (standard)	0.0%	7.2%	36.4%	60.5%	65.6%	+6.2
Dense retrieval (MiniLM)	99.7%	7.2%	36.4%	59.3%	64.8%	+5.4
Cluster-shared retrieval	99.6%	7.0%	35.0%	59.4%	64.1%	+4.7
Structural retrieval	99.7%	7.2%	34.8%	58.2%	62.5%	+3.1

to MEM-NOREPLAY (59.4%, $\approx$ base), incurring no penalty when memory cannot engage. Thus Mem-MaxRL is a strict, safe generalization of smoothed MAXRL.

Non-recurring streams via semantic retrieval. The exact-match buffer is inert when prompts never recur (rescue rate 0.0% at $N_{\text{prompts}}=\infty$). To show the deadlock-rescue *mechanism* is not tied to exact keys, we replace the per-prompt lookup with a global success buffer $\mathcal{M}_{\text{global}}$ queried by retrieval, and test on a fully non-recurring stream of 25,600 unique prompts (64 brand-new prompts per step, none ever repeated). We compare (i) exact-match, (ii) *dense* retrieval embedding the prompt with all-MiniLM-L6-v2 (Reimers & Gurevych, 2019) and taking the cosine-nearest verified trajectory above $\tau=0.85$, (iii) *cluster-shared* retrieval over $K=16$ online centroids, and (iv) *structural* retrieval on grid representations. Crucially, because a retrieved z^* was produced for a *different* historical prompt x' , it may be invalid for the current prompt x ; we therefore gate every cross-prompt replay behind a **target-prompt re-verification** filter $f(z^*, x)=y^*(x)$ that re-runs the deterministic reward verifier on x before the update fires (Section 5.3). As Table 3 and Figure 10 show, dense retrieval raises the neighbor-lookup rate from 0.0% to **99.7%** (14,979/15,025 deadlocks matched), and—once gated by re-verification—all retrieval variants keep coverage above the pre-RL base (62.5%–64.8% vs. 59.4%). Thus semantic memory is a viable, safe extension of Mem-MaxRL to the dynamic non-recurring prompt streams typical of large-scale post-training.

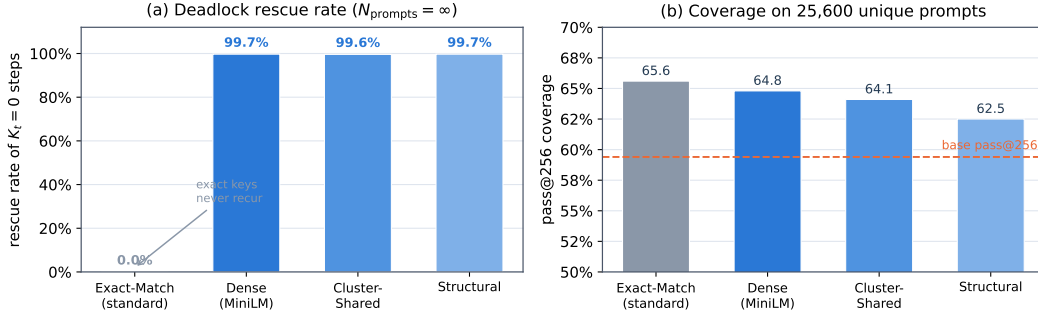


Figure 10: Non-recurring stream (25,600 unique prompts, $N_{\text{prompts}} = \infty$). (a) Exact-match memory rescues 0% of $K_t=0$ deadlocks, while dense/cluster/structural retrieval restore rescue to $\approx 99.7\%$. (b) All retrieval variants keep pass@256 coverage above the pre-RL base (dashed), making semantic memory a safe extension to dynamic prompt streams.

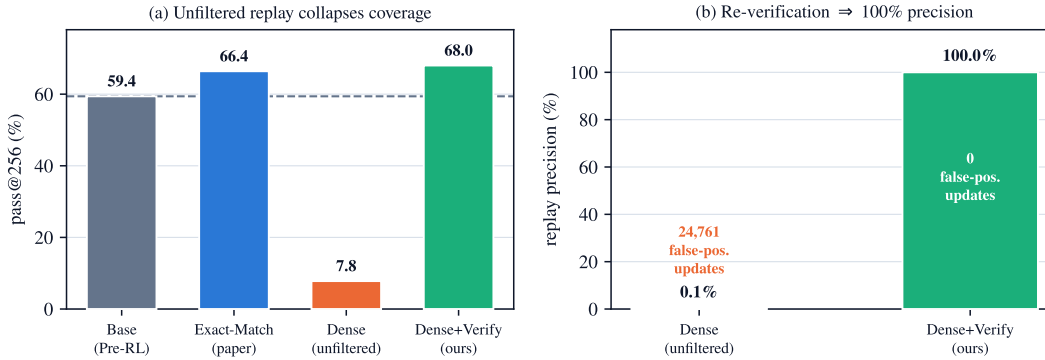


Figure 11: Target-prompt re-verification on the non-recurring 25,600-maze stream. (a) Unfiltered cross-prompt replay collapses pass@256 to 7.8% (below the pre-RL base, dashed), whereas re-verified dense retrieval reaches 68.0%—matching exact-match memory. (b) Re-verification lifts replay precision from 0.1% to 100.0%, eliminating all 24,761 false-positive updates.

Target-prompt re-verification is essential. Semantically similar prompts (e.g. mazes with different start/goal cells) have *different* correct solutions, so replaying a retrieved z^* *without* checking it against x injects false-positive gradients. Figure 11 isolates this on the 25,600-maze stream. *Unfiltered* dense retrieval fires on 99.8% of deadlocks, but only 0.14% of retrieved completions solve the target maze; the 24,761 false-positive updates collapse pass@256 to 7.8% (−51.6 vs. base). Gating replay behind the verifier $f(z^*, x) = y^*(x)$ raises precision to 100.0% (*zero* false positives)—when z^* fails on x , $\mathcal{L}_{\text{replay}} = 0$ and the step reduces to the smoothed MAXRL floor—so re-verified retrieval reaches 68.0% pass@256 (+8.6 over base). The verifier is the same deterministic reward used for on-policy rollouts (< 1% of a forward pass), a cheap, indispensable guardrail for cross-prompt memory (Appendix E).

6 CONCLUSION

We introduced Mem-MaxRL, a memory-augmented maximum-likelihood RL method that prevents the coverage collapse of expected-reward RL and repairs the finite-sample failure modes of MAXRL. Pairing an EMA-smoothed likelihood weight with a per-prompt Success Buffer whose replays target the conditional ML gradient converts wasted $K_t=0$ deadlocks into productive updates. On a 17×17 maze it reaches 66.4% pass@256—the only method to exceed the base model—beating direct pass@ k optimizers (PKPO, Unlikelihood Reward), coverage (SimKO), and replay (RLEP) baselines; it scales to Qwen2.5-3B on GSM8K/MATH-500 with the highest diversity gap and stable 300-step training, and a hybrid objective recovers pass@1. When prompts do not recur, exact-match memory degrades gracefully to smoothed MAXRL, and re-verified dense retrieval safely rescues deadlocks on a 25,600-prompt non-recurring stream. **Limitations and future work.** Replay benefits from prompt recurrence or retrieval; extending Mem-MaxRL to non-binary/dense rewards, larger (7B+) models, and code / theorem-proving domains are promising directions.

REFERENCES

- Sanghwan Bae, Jiwoo Hong, M. Lee, Hanbyul Kim, JeongYeon Nam, and Donghyun Kwak. Online difficulty filtering for reasoning oriented reinforcement learning. In *Conference of the European Chapter of the Association for Computational Linguistics*, pp. 700–719, 2025.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Pondé, Jared Kaplan, Harrison Edwards, Yura Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, G. Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mo Bavarian, Clemens Winter, P. Tillet, F. Such, D. Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William H. Guss, Alex Nichol, Igor Babuschkin, S. Balaji, Shantanu Jain, A. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, M. Knight, Miles Brundage, M. Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, I. Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. *ArXiv*, abs/2107.03374, 2021.
- Zhipeng Chen, Xiaobo Qin, Youbin Wu, Yue Ling, Qinghao Ye, Wayne Xin Zhao, and Guang Shi. Pass@k training for adaptively balancing exploration and exploitation of large reasoning models. *ArXiv*, abs/2508.10751, 2025.
- Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jian-shu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for llm reasoning from a cross-domain perspective. *ArXiv*, abs/2506.14965, 2025.
- K. Cobbe, Vineet Kosaraju, Mo Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *ArXiv*, abs/2110.14168, 2021.
- L. Espeholt, Hubert Soyer, R. Munos, K. Simonyan, Volodymyr Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, S. Legg, and K. Kavukcuoglu. Impala: Scalable distributed deep-rl with importance weighted actor-learner architectures. *ArXiv*, abs/1802.01561, 2018.
- Jingchu Gai, Guanning Zeng, Huaqing Zhang, and Aditi Raghunathan. Differential smoothing mitigates sharpening and improves llm reasoning. *ArXiv*, abs/2511.19942, 2025.
- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, pp. 10835–10866, 2022.
- Caglar Gulcehre, T. Paine, S. Srinivasan, Ksenia Konyushkova, L. Weerts, Abhishek Sharma, Aditya Siddhant, Alexa Ahern, Miaosen Wang, Chenjie Gu, Wolfgang Macherey, A. Doucet, Orhan Firat, and Nando de Freitas. Reinforced self-training (rest) for language modeling. *ArXiv*, abs/2308.08998, 2023.
- Anthony GX-Chen, Jatin Prakash, Jeff Guo, Rob Fergus, and R. Ranganath. Kl-regularized reinforcement learning is designed to mode collapse. *ArXiv*, abs/2510.20817, 2025.
- Andren He, Daniel Fried, and S. Welleck. Rewarding the unlikely: Lifting grpo beyond distribution sharpening. In *Conference on Empirical Methods in Natural Language Processing*, pp. 25548–25560, 2025.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, D. Song, and J. Steinhardt. Measuring mathematical problem solving with the math dataset. *ArXiv*, abs/2103.03874, 2021.
- Arian Hosseini, Xingdi Yuan, Nikolay Malkin, Aaron C. Courville, Alessandro Sordoni, and Rishabh Agarwal. V-star: Training verifiers for self-taught reasoners. *ArXiv*, abs/2402.06457, 2024.

- Miaobo Hu, Shuhao Hu, Bokun Wang, Ruohan Wang, Xin Wang, Xiaobo Guo, Daren Zha, and Jun Xiao. Agpo: Adaptive group policy optimization with dual statistical feedback. *ArXiv*, abs/2605.20722, 2026.
- Xiaotong Ji, Rasul Tutunov, Matthieu Zimmer, and Haitham Bou-Ammar. Scalable power sampling: Unlocking efficient, training-free reasoning for llms via distribution sharpening. *ArXiv*, abs/2601.21590, 2026.
- Jean Kaddour. Target policy optimization. *ArXiv*, abs/2604.06159, 2026.
- Youngeun Kim. Mc-grpo: Median-centered group relative policy optimization for small-rollout reinforcement learning. *ArXiv*, abs/2601.22582, 2026.
- Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and R. Raileanu. Understanding the effects of rlhf on llm generalisation and diversity. *ArXiv*, abs/2310.06452, 2023.
- Germán Kruszewski, Pierre Erbacher, Jos Rozen, and Marc Dymetman. Whatever remains must be true: Filtering drives reasoning in llms, shaping diversity. *ArXiv*, abs/2512.05962, 2025.
- Long Li, Jiaran Hao, J. Liu, Zhijian Zhou, Xiaoyu Tan, Wei Chu, Zhe Wang, Shirui Pan, Chao Qu, and Yuan Qi. The choice of divergence: A neglected key to mitigating diversity collapse in reinforcement learning with verifiable reward. *ArXiv*, abs/2509.07430, 2025.
- Chen Liang, Mohammad Norouzi, Jonathan Berant, Quoc V. Le, and N. Lao. Memory augmented policy optimization for program synthesis and semantic parsing. In *Neural Information Processing Systems*, pp. 10015–10027, 2018.
- Xiao Liang, Zhong zhi Li, Yeyun Gong, Yelong Shen, Yingchun Wu, Zhijiang Guo, and Weizhu Chen. Beyond pass@1: Self-play with variational problem synthesis sustains rlvr. *ArXiv*, abs/2508.14029, 2025.
- Zichuan Liu, Jinyu Wang, Lei Song, and Jiang Bian. Sample-efficient llm optimization with reset replay. *ArXiv*, abs/2508.06412, 2025.
- Zijun Min, Bingshuai Liu, Ante Wang, Long Zhang, Anxiang Zeng, Haibo Zhang, and Jinsong Su. Orchestrating tokens and sequences: Dynamic hybrid policy optimization for rlvr. *ArXiv*, abs/2601.05607, 2026.
- Ruotian Peng, Yi Ren, Zhouliang Yu, Weiyang Liu, and Yandong Wen. Simko: Simple pass@k policy optimization. *ArXiv*, abs/2510.14807, 2025.
- Xue Bin Peng, Aviral Kumar, Grace Zhang, and S. Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. *ArXiv*, abs/1910.00177, 2019.
- A. Plaatt, Annie Wong, Suzan Verberne, J. Broekens, Niki van Stein, and T. Bäck. Multi-step reasoning with large language models, a survey. *ACM Computing Surveys*, 58:1 – 35, 2024.
- Rafael Rafailov, Archit Sharma, E. Mitchell, S. Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *ArXiv*, abs/2305.18290, 2023.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *EMNLP*, 2019.
- Amrith Rajagopal Setlur, Nived Rajaraman, Sergey Levine, and Aviral Kumar. Scaling test-time compute without verification or rl is suboptimal. *ArXiv*, abs/2502.12118, 2025.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, R. Xu, Jun-Mei Song, Mingchuan Zhang, Y. K. Li, Yu Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *ArXiv*, abs/2402.03300, 2024.
- Maojiang Su, Po-Chung Hsieh, Weimin Wu, Mingcheng Lu, Jiun-Man Chen, Jerry Yao-Chieh Hu, and Han Liu. Discrete flow matching policy optimization. *ArXiv*, abs/2604.06491, 2026.

- Ke Sun, Yizhou Zhao, J. Xin, Qi Long, and Weijie J. Su. Curverl: Principled distribution-aware context reweighting for llm reasoning. *ArXiv*, abs/2605.24331, 2026.
- Fahim Tajwar, Guanning Zeng, Yue Zhou, Yuda Song, Daman Arora, Yiding Jiang, Jeff G. Schneider, Ruslan Salakhutdinov, Haiwen Feng, and Andrea Zanette. Maximum likelihood reinforcement learning. *ArXiv*, abs/2602.02710, 2026.
- Qwen Team. Qwen2.5 technical report. *ArXiv*, abs/2412.15115, 2024.
- Christos Thrampoulidis, Sadegh Mahdavi, and Wenlong Deng. Advantage shaping as surrogate reward maximization: Unifying pass@k policy gradients. *Trans. Mach. Learn. Res.*, 2026, 2025.
- Hugo Touvron, Louis Martin, Kevin R. Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bash-lykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, D. Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, A. Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel M. Kloumann, A. Korenev, Punit Singh Koura, M. Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, X. Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, J. Reizenstein, Rashi Rungta, Kalyan Saladi, A. Schelten, Ruan Silva, Eric Michael Smith, R. Subramanian, Xia Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zhengxu Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, M. Kambadur, Sharan Narang, Aur’elien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models. *ArXiv*, abs/2307.09288, 2023.
- Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, L. Wang, Antonia Creswell, G. Irving, and I. Higgins. Solving math word problems with process- and outcome-based feedback. *ArXiv*, abs/2211.14275, 2022.
- Christian J. Walder and Deep Karkhanis. Pass@k policy optimization: Solving harder reinforcement learning problems. *ArXiv*, abs/2505.15201, 2025.
- Chaoqi Wang, Yibo Jiang, Chenghao Yang, Han Liu, and Yuxin Chen. Beyond reverse kl: Generalizing direct preference optimization with diverse divergence constraints. *ArXiv*, abs/2309.16240, 2023.
- Xiaoxuan Wang, Bo Liu, Song Jiang, Jingzhou Liu, Jingyuan Qi, Xia Chen, and Baosheng He. From solving to verifying: A unified objective for robust reasoning in llms. *ArXiv*, abs/2511.15137, 2025.
- Wei Xiong, Jiarui Yao, Yuhui Xu, Bo Pang, Lei Wang, Doyen Sahoo, Junnan Li, Nan Jiang, Tong Zhang, Caiming Xiong, and Hanze Dong. A minimalist approach to llm reasoning: from rejection sampling to reinforce. *ArXiv*, abs/2504.11343, 2025.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxin Yang, Jingren Zhou, Jingren Zhou, Junyan Lin, K. Dang, Keqin Bao, Ke-Pei Yang, Le Yu, Li-Chun Deng, Mei Li, Min Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yi-Chao Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. In *arXiv*, 2025.
- E. Zelikman, Yuhuai Wu, and Noah D. Goodman. Star: Bootstrapping reasoning with reasoning. In *arXiv*, 2022.
- Dan Zhang, Sining Zhou, Yisong Yue, Yuxiao Dong, and Jie Tang. Rest-mcts*: Llm self-training via process reward guided tree search. *ArXiv*, abs/2406.03816, 2024.
- Hongzhi Zhang, Jia Fu, Jingyuan Zhang, Kai Fu, Qi Wang, Fuzheng Zhang, and Guorui Zhou. Rlep: Reinforcement learning with experience replay for llm reasoning. *ArXiv*, abs/2507.07451, 2025a.

Kai Zhang, Yuzhong Hong, Junwei Bao, Hongfei Jiang, Yang Song, Dingqian Hong, and Hui Xiong. Gvpo: Group variance policy optimization for large language model post-training. *ArXiv*, abs/2504.19599, 2025b.

A THEORETICAL PROOFS

A.1 PROOF OF THE CONDITIONAL ML GRADIENT (EQ. 2)

Proof. Let $p_\theta(y^*(x) | x) = \sum_{z \in \mathcal{Z}} m_\theta(z | x) \mathbb{I}\{f(z) = y^*(x)\}$. Differentiating $J_{\text{ML}}(\theta) = \log p_\theta(y^*(x) | x)$,

$$\nabla_\theta J_{\text{ML}}(\theta) = \frac{\nabla_\theta p_\theta(y^*(x) | x)}{p_\theta(y^*(x) | x)} = \frac{1}{p_\theta(y^*(x) | x)} \sum_z \nabla_\theta m_\theta(z | x) \mathbb{I}\{f(z) = y^*(x)\} \quad (7)$$

$$= \sum_z \frac{m_\theta(z | x) \mathbb{I}\{f(z) = y^*(x)\}}{p_\theta(y^*(x) | x)} \nabla_\theta \log m_\theta(z | x) \quad (8)$$

$$= \sum_z m_\theta(z | x, f(z) = y^*(x)) \nabla_\theta \log m_\theta(z | x) = \mathbb{E}[\nabla_\theta \log m_\theta(z | x) | f(z) = y^*(x)]. \quad (9)$$

Thus the ML gradient equals the expected score over successful trajectories. $\square$

A.2 PROOF OF ESTIMATOR-OBJECTIVE EQUIVALENCE

Proof. With N i.i.d. rollouts, $r_i = \mathbb{I}\{f(z_i) = y^*(x)\}$, $K = \sum_i r_i$, the unnormalized-baseline estimator is $\hat{g}_N(x) = \frac{1}{K} \sum_i r_i \nabla_\theta \log m_\theta(z_i | x)$ for $K \geq 1$ and 0 otherwise. Taking expectations,

$$\mathbb{E}[\hat{g}_N(x)] = \sum_{k=1}^N P(K = k) \mathbb{E}\left[\frac{1}{k} \sum_i r_i \nabla_\theta \log m_\theta(z_i | x) \mid K = k\right] \quad (10)$$

$$= \left(\sum_{k=1}^N P(K = k)\right) \nabla_\theta J_{\text{ML}}(x) = (1 - (1 - p)^N) \nabla_\theta J_{\text{ML}}(x) = \nabla_\theta J_{\text{MAXRL}}^{(N)}(x). \quad (11)$$

Hence $\hat{g}_N(x)$ is unbiased for the order- N truncated ML gradient. The control-variate $V_N = \frac{1}{N} \sum_i \nabla_\theta \log m_\theta(z_i | x)$ has $\mathbb{E}[V_N] = 0$, so subtracting it (Eq. 3) preserves unbiasedness while reducing variance. $\square$

B AUTOGRAD SEMANTICS OF THE REPLAY GRADIENT (EQ. 5)

We give the full derivation underlying Eq. 6. On a deadlock ($K_t = 0$) with a replayed success $z^* \sim \mathcal{M}(x)$, the implemented loss applies a stop-gradient to the entire advantage weight, so that in autograd

$$\mathcal{L}_{\text{replay}}(\theta; x) = - \underbrace{\left[w(x) \cdot \text{clip}(r_\theta(z^*), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \right]}_{A_{\text{replay}}^*(x) \in \mathbb{R}} \log m_\theta(z^* | x), \quad r_\theta(z^*) = \frac{m_\theta(z^* | x)}{m_{\text{old}}(z^* | x)}, \quad (12)$$

where $[\cdot]_\perp$ denotes $\text{detach}()$ (stop-gradient). Since $A_{\text{replay}}^*(x)$ carries no gradient, differentiation touches only the score term:

$$\nabla_\theta \mathcal{L}_{\text{replay}}(\theta; x) = -A_{\text{replay}}^*(x) \nabla_\theta \log m_\theta(z^* | x). \quad (13)$$

Had the ratio $r_\theta(z^*)$ instead been left attached, the product rule would have added a non-canonical term $-w(x) \nabla_\theta r_\theta(z^*) \log m_\theta(z^* | x)$; the detachment removes it by construction. The resulting update is a single positive, weighted score-function step, i.e. exactly one Monte-Carlo sample of the conditional ML gradient $\mathbb{E}[\nabla_\theta \log m_\theta(z | x) | f(z) = y^*(x)]$ (Eq. 2), with $A_{\text{replay}}^*(x)$ serving as a PPO trust-region weight that keeps the off-policy correction bounded. This is the standard REINFORCE-with-detached-advantage estimator used by GRPO/PPO surrogates, not a new objective.

Table 4: Offset ϵ sweep at $\eta = 0.3$ (recurring maze, base pass@256 = 60.2%).

ϵ	pass@1	pass@16	pass@128	pass@256	mean grad-norm
10^{-6}	3.4%	28.9%	60.1%	68.0%	0.429
10^{-5}	3.6%	31.2%	61.7%	69.5%	0.428
10^{-3}	3.8%	30.5%	57.0%	64.1%	0.321
10^{-1}	4.0%	29.7%	56.5%	64.1%	0.047

Table 5: Hybrid objective sweep on Qwen2.5-3B-Instruct ($k=64$ samples/prompt). α is the Mem-MaxRL mixing weight ($\alpha=0$: pure GRPO; $\alpha=1$: pure Mem-MaxRL). Best per column in **bold**.

α	GSM8K (in-distribution)				MATH-500 (out-of-distribution)		
	pass@1	pass@16	pass@64	Div. gap	pass@1	pass@64	Div. gap
0.0 (pure GRPO)	71.0%	90.4%	95.0%	24.0	48.8%	82.5%	33.8
0.5	78.5%	92.0%	97.5%	19.0	52.7%	85.0%	32.3
0.7	78.6%	94.4%	100.0%	21.4	51.8%	85.0%	33.2
0.9	72.5%	94.4%	100.0%	27.5	43.2%	75.0%	31.8
1.0 (pure Mem-MaxRL)	75.0%	93.2%	97.5%	22.5	46.4%	87.5%	41.1

Table 6: Target-prompt re-verification on the non-recurring 25,600-maze stream. Re-verification converts unsafe cross-prompt replay (pass@256 = 7.8%) into the best-coverage variant (68.0%) with zero false-positive updates.

Strategy	Retrieval	Replay prec.	False-pos.	pass@1	pass@16	pass@256	Δ pass@256
Base model (Pre-RL)	—	—	—	1.1%	13.4%	59.4%	—
Exact-Match (paper standard)	0.0%	—	0	7.6%	37.0%	66.4%	+7.0
Dense retrieval (unfiltered)	99.8%	0.1%	24,761	0.1%	1.1%	7.8%	−51.6
Dense + target-verify (ours)	99.7%	100.0%	0	5.9%	35.8%	68.0%	+8.6

C OFFSET ϵ SWEEP

Table 4 sweeps the stabilization offset ϵ at fixed $\eta = 0.3$. Small offsets (10^{-6} – 10^{-5}) are indistinguishable and best; a large $\epsilon = 10^{-1}$ dominates the denominator for hard (small- $\bar{p}$) prompts, flattening the $1/p$ maximum-likelihood curve (mean weight batch-variance drops from 104.7 to 4.6) and costing ~ 3.9 points of pass@256—the attenuation-of- $1/p$ failure mode.

D HYBRID OBJECTIVE INTERPOLATION (FULL NUMBERS)

Table 5 reports the full sweep behind Figure 6(b). The hybrid loss $\mathcal{L}_{\text{hybrid}} = \alpha \mathcal{L}_{\text{Mem-MaxRL}} + (1 - \alpha) \mathcal{L}_{\text{GRPO}}$ is computed over the same rollout batch on Qwen2.5-3B-Instruct. Setting $\alpha=0.7$ maximizes GSM8K pass@1 (78.6%) while attaining 100.0% pass@64 coverage and retaining strong out-of-distribution MATH-500 coverage (85.0% pass@64), tracing a favorable point on the pass@1-vs-coverage Pareto frontier.

E TARGET-PROMPT RE-VERIFICATION (FULL BREAKDOWN)

Table 6 gives the complete metrics behind Figure 11 on the fully non-recurring stream of 25,600 unique 17×17 mazes (400 steps $\times$ 64 new mazes/step). “Retrieval rate” is the fraction of $K_t=0$ deadlocks for which a neighbor above $\tau=0.85$ exists; “replay precision” is the fraction of *fired* replays whose z^* actually solves the target maze x . Unfiltered dense retrieval fires almost always but is 99.86% wrong, injecting 24,761 false-positive updates that collapse coverage. Gating replay behind $f(z^*, x)=y^*(x)$ eliminates every false positive (precision 100.0%) and yields the best coverage (68.0% pass@256).

F FULL HYPERPARAMETERS

Table 7: Hyperparameter configuration for the maze reasoning experiments.

Hyperparameter	Value
Model	Qwen2-style transformer (3M params)
Layers / hidden / heads	4 / 256 / 4
Vocab / max sequence length	32 / 512 tokens
Training / eval mazes	256 recurring / 128 held-out
Batch size / rollouts N	64 / 32 (2,048 samples/step)
Steps / learning rate	400 / 5×10^{-5} (AdamW)
EMA smoothing η	0.3 (swept $\{0.05, 0.1, 0.3, 0.5, 1.0\}$)
Offset ϵ	10^{-6} (swept $\{10^{-5}, 10^{-3}, 10^{-1}\}$)
Buffer capacity $M_{\max}$	4 completions per prompt
Replay clip ϵ_{clip}	0.2
Evaluation sampling	256 combinatorial samples/maze (temp 1.0)